DON BOSCO INSTITUTE OF TECHNOLOGY BENGALURU

Department of CSE (AI & ML)

Academic year: 2026-27 (ODD)

Major Project Phase-II (BCI786) (7th Semester)

Project Title: EchoSphere - A Secure AI- and IoT-Enabled Framework for Smart Campus Communication and Announcement Automation

Abstract: EchoSphere is a centralized smart campus announcement management platform designed to replace physical notice boards with role-based access control (RBAC) across 6 hierarchical roles (DevAdmin, Principal, College Admin, HoD, Teacher, Student), automated multi-tier approval workflows, and multi-channel real-time broadcasting. The system integrates an AI microservice engine (utilizing Google Gemini 1.5 and Whisper STT for notice drafting, summarization, spam filtering, duplicate detection, priority/category tagging, and voice synthesis) with real-time WebSocket feeds, FCM mobile push notifications, and a planned IoT smart speaker hardware module (ESP32-S3 / Raspberry Pi Zero 2 W with PCM5102A I2S DAC and TPA3116D2 amplifier). The cross-platform Flutter application (deployed for Android and Windows) ensures zero-layout visual overflow and maintains an immutable database audit log.

Keywords: Artificial Intelligence, Smart Campus, Multi-Tier Approval, Role-Based Access Control, IoT Speaker Nodes, Voice Synthesis, Flutter, FastAPI, PostgreSQL.

Problem Statement:

Traditional campus communication relies on physical notice boards, manual public address (PA) announcements, scattered messaging groups, and paper circulars, leading to communication delays, missed notices, lack of approval verification, spamming, and an inability to deliver urgent emergency announcements instantaneously to both mobile devices and physical campus locations. Existing platforms lack automated AI content validation, priority triage, and integrated hardware speaker queuing. EchoSphere addresses these challenges by providing a secure, AI-assisted framework that automates notice creation, enforces hierarchical multi-tier approvals, schedules broadcasts, and delivers real-time notifications across mobile feeds, push alerts, and physical IoT speakers.

Objectives:

1. Develop a multi-tiered smart campus announcement platform enforcing 6-role hierarchical RBAC (DevAdmin, Principal, College Admin, HoD, Teacher, Student).
2. Enforce structured approval workflows (Draft -> Pending Approval -> Approved / Rejected / Scheduled / Archived) with immutable audit logs.
3. Integrate an AI microservice engine (Google Gemini 1.5 & Whisper STT) for notice drafting, expansion, spam filtering, duplicate detection, and voice synthesis.
4. Architect departmental IoT smart speaker hardware nodes (ESP32-S3 / Raspberry Pi Zero 2 W with PCM5102A DAC, TPA3116D2 amplifier, and MQTT topic queues) for public broadcasts.
5. Enable real-time multi-channel delivery via WebSockets (in-app feeds), Firebase Cloud Messaging (FCM push notifications), and zero-layout overflow Android/Windows apps.

Outcome:

EchoSphere provides an adaptive, secure campus announcement platform combining AI notice enhancement, multi-tier approval workflows, multi-channel real-time delivery, hardware smart speaker architecture, and security audit logging. The system eliminates manual communication bottlenecks and improves the timeliness, accuracy, and reach of academic and emergency announcements.

Language / Software Tools used:

OS & Clients: Windows; Android (Mobile/Tablet), Windows (Desktop) | **Languages:** Python 3.11+, Dart 3.x | **Frameworks:** Flutter 3.x, FastAPI (Uvicorn, Async SQLAlchemy, Pydantic, Alembic) | **Database & Cache:** PostgreSQL 15, Redis | **AI/ML & Voice Stack:** Google Gemini 1.5 API, Whisper STT, PyTorch, Neural TTS | **Hardware Module (Planned):** ESP32-S3 / Raspberry Pi Zero 2 W, PCM5102A I2S DAC, TPA3116D2 Class-D Amplifier, 5V Relay Module, PA / Ceiling Speakers | **APIs & Protocols:** REST API, WebSockets, MQTT, FCM

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